Implementation and Numerical Diagnostics of Row-Splitting ILU Preconditioners

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Abstract

This report presents a Python implementation and numerical investigation of the row-splitting incomplete LU (ILU) preconditioner of Scott and Tůma. A common baseline was evaluated on 22 sparse matrices and compared with the published dense-reduced-system results. The study combines factorization traces, triangular-solve diagnostics, and preconditioned CGLS behavior, rather than relying on iteration counts alone. Two findings are examined in detail: extreme sensitivity of an upper triangular factor in Problem 5, which is consistent with the accuracy limitations discussed in the source paper, and a discrepancy between the printed CGLS initialization and its standard preconditioned recurrence. The experiments also show that additional ILU fill does not necessarily reduce observed triangular-solve amplification. Unresolved implementation differences are distinguished from established numerical observations.

1. Method and computational implementation

For a row- and column-permuted rectangular matrix, the approximate factorization is

$$P_r A P_c \approx \begin{bmatrix} \widetilde{L}_1 \\ \widetilde{L}_2 \end{bmatrix} \widetilde{U}, \qquad \widetilde{Y} = \widetilde{L}_2 \widetilde{L}_1^{-1}, \qquad \widetilde{S} = I + \widetilde{Y}\widetilde{Y}^\top. \tag{1}$$

The column-oriented ILUP(p, τ) factorization controls retained fill and dropping, with threshold pivoting (μ) and small-pivot modification. A Woodbury-type correction then yields the preconditioner action through triangular solves and a reduced system. In the permuted block coordinates, its core steps are

$$u = r_2 - \widetilde{Y} r_1, \quad \widetilde{S} w = u, \quad y = r_1 + \widetilde{Y}^\top w, \quad \widetilde{L}_1 v = y, \quad \widetilde{U} h = v. \tag{2}$$

The resulting h, after the corresponding permutation, is the preconditioned CGLS gradient. The implementation includes sparse elimination, logical row swaps, dropping, pivot modification, and matrix-free applications of $\widetilde{Y}$. Dense reference and operator-level checks were used for small systems. The reported benchmark baseline explicitly constructs and solves the dense $\widetilde{S}$ system, matching the treatment in the source paper's Table 2 [1].

2. Benchmark design and 22-problem comparison

The baseline uses $p = 10$, $\tau = 0$, $\mu = 0.1$, and small $= 10^{-10}$, with zero initial guess, a 2,000-iteration CGLS cap, and the Papež–Tichý stopping estimator ratio$_{\mathrm{PT}} \le 10^{-10}$. The matrices were preprocessed by removing empty rows/columns, transposing when required, and normalizing the nonzero columns. The available runs cover 22 of the paper's 24 test matrices; Problems 14 (`LARGE001`) and 15 (`DELF000`) were excluded because the corresponding source matrices were not available in the local dataset.

Table 1. Published iteration counts and local CGLS termination ($\tau = 0$) for 22 benchmark matrices.

No.	Paper its	Local termination	No.	Paper its	Local termination
1	2	Breakdown (0)	12	279	Converged
2	3	Converged	13	†	Limit (2000)
3	3	Breakdown	16	10	Breakdown
4	3	Breakdown	17	8	Breakdown
5	†	Breakdown (2)	18	6	Breakdown
6	4	Converged	19	†	Converged (19)
7	2	Breakdown	20	29	Breakdown
8	3	Breakdown	21	43	Converged
9	10	Converged	22	3	Breakdown
10	36	Breakdown	23	21	Breakdown
11	10	Converged	24	19	Converged

Notes. †: requested tolerance not reached in the original paper. "Breakdown" denotes a local arithmetic/recurrence termination, not an iteration-limit exit. These categories are not claims of identical numerical trajectories.

Eight local runs were recorded as converged, thirteen as breakdown, and one reached the iteration cap. Thirteen matrices differ from the paper in the qualitative termination outcome (twelve local breakdowns against reported convergence, plus the reverse discrepancy in Problem 19). Thus, matching broad behavior in some matrices should not be presented as full numerical reproduction. In particular, modified-pivot counts differed in Problems 1 (289 locally versus 163 published), 3 (83 versus 162), and 8 (0 versus 9), pointing to potential divergence before the outer CGLS iteration.

3. Numerical diagnostics and interpretation

3.1. Problem 1: a factorization-path discrepancy

The local triangular solve produced an amplification of about 5.89×10^{32} and CGLS broke down at initialization, whereas the paper reports two iterations. These outcomes refer to *different computed factors*: the modified-pivot counts already differ (289 versus 163). The local amplification therefore does not contradict the paper's fast convergence; it is evidence that the preconditioners may not be numerically equivalent. Specific candidates for a controlled comparison include the remaining-row-count update and tie handling in Algorithm 5, the original-versus-updated column norm used by the modification rule in Algorithm 6, and the ordering of dropping relative to column scaling. These are hypotheses, not identified implementation defects. A discriminating experiment would freeze the input, trace the first divergent pivot/row count, and test these choices separately while preserving the baseline output.

3.2. Problem 5: linking factor sensitivity to the reported accuracy plateau

The estimated upper-triangular condition number and observed amplification were

$$\kappa_1(\widetilde{U}) \approx 2.15 \times 10^{38}, \qquad \frac{\|h\|_2}{\|v\|_2} \approx 1.27 \times 10^{32}, \qquad \widetilde{U}h = v. \tag{3}$$

The `spsolve_triangular` solution had normalized backward error

$$\eta_\infty = \frac{\|v - \widetilde{U}h\|_\infty}{\|\widetilde{U}\|_\infty \|h\|_\infty + \|v\|_\infty} \approx 3.25 \times 10^{-20}, \tag{4}$$

but two independent solve paths differed by about 0.2475 in relative norm. This is an illustration of standard numerical conditioning theory, *not* a new backward-stability result. Importantly, the paper attributes the slowly decreasing and nearly stationary ratio_{PT} curves for Problems 5 and 13 in its Figure 2 to an approximately reached attainable-accuracy limit [1]. The locally observed ill-conditioning supplies a plausible *mechanism compatible with* this explanation for Problem 5. It does not establish that the authors' $\widetilde{U}$ has the same condition number or reproduce their stagnation trajectory: the local Problem 5 run stopped by breakdown after two iterations.

3.3. Problems 13 and 19: different mechanisms

Problem 13 had sampled solve amplification 55.5, factor growth 8.75, and no modified pivots. The local CGLS run reached 2,000 iterations with $\text{ratio}_{\text{PT}} \approx 3.65 \times 10^{-9}$, compared with the paper's 2.50×10^{-5}. It does not show the extreme triangular amplification seen in Problem 5. Conversely, Problem 19 locally converged in 19 iterations ($\text{ratio}_{\text{PT}} \approx 7.27 \times 10^{-11}$), while the paper reported † (1.42×10^{-8}). Preconditioner quality, factorization details, and floating-point recurrence behavior must therefore be analyzed separately.

3.4. Sensitivity to the ILUP fill parameter

Table 2 summarizes the saved one-factor-at-a-time parameter sweep for Problem 1. In addition to factor fill and amplification, it reports local iteration counts and recorded CGLS termination flags. The archived sweep summary does not provide a per-configuration ratio_{PT} trace, so no such values are inferred.

Table 2. Problem 1, varying p with other parameters fixed at the baseline.

p	Relative fill	Sampled amplification	Local CGLS exit (its)
2	1.16	4.41×10^{2}	Flagged convergence (4)
5	1.64	2.57×10^{10}	Breakdown (0)
8	2.05	2.12×10^{28}	Breakdown (0)
10	2.31	5.89×10^{32}	Breakdown (0)
15	2.93	1.31×10^{36}	Breakdown (0)
20	3.54	1.65×10^{32}	Breakdown (1)
40	5.62	4.58×10^{35}	Breakdown (1)

Note. "Flagged convergence" refers to the saved solver indicator; independent accuracy is not asserted. The amplification is evaluated for a run-specific right-hand side, not as $\|\widetilde{U}^{-1}\|$.

The strongly non-monotone sampled amplification cautions against using fill alone as a stability surrogate. A controlled follow-up should use common probe vectors and Hager–Higham-type inverse-norm or condition estimation across all p, alongside final ratio_{PT}, normal residuals, and iteration histories; these additional measurements have *not* been performed for the reported sweep.

4. Consistency check of preconditioned CGLS

Algorithm 1 in the source paper prints $p_0 = z_0$, whereas the local implementation sets $p_0 = h_0 = M^{-1}z_0$ with $z_0 = A^\top r_0$. For the printed step length and recurrence,

$$\alpha_0 = \frac{z_0^\top h_0}{\|Ap_0\|_2^2}, \qquad z_1 = z_0 - \alpha_0 A^\top A p_0, \tag{5}$$

choosing $p_0 = h_0$ gives $z_1^\top p_0 = 0$, while $p_0 = z_0$ generally does not. An independent 3×2 example yielded $\|A^\top r_2\|_2 \approx 2.22 \times 10^{-16}$ after two iterations with $p_0 = h_0$, versus 0.539 with $p_0 = z_0$ and the rest of the recurrence unchanged. This supports the local choice as consistent with standard preconditioned CGLS; the printed initialization likely contains a notational error, although the authors have not confirmed it. This mathematical check does not establish correctness of every local implementation detail.

5. Discussion and conclusion

The most useful methodological outcome is to separate (i) elimination, pivot, and fill diagnostics; (ii) triangular-solve amplification and backward error; and (iii) outer Krylov residuals, error estimates, and exit conditions. The Problem 5 results provide a conditioning-based interpretation compatible with the paper's Figure 2, while Problems 1, 13, and 19 exhibit different sources of numerical discrepancy. The local data also motivate a factor-aware stability screen, but this remains a diagnostic proposal rather than a validated new preconditioner.

An especially relevant next comparison is the paper's Table 3 observation that replacing the dense reduced solve by two CG steps, or even using $\widetilde{S} \approx I$, can outperform dense factorization [1]. Such a comparison would help separate inaccuracies in the ILU factors from effects of the reduced correction. These alternatives are not presented here as completed experiments.

References

[1] J. A. Scott and M. Tůma, "Row-Splitting ILU Preconditioners for Sparse Least-Squares Problems," *arXiv:2603.28642*, 2026. https://arxiv.org/abs/2603.28642.